



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to M/S MAGIZH FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

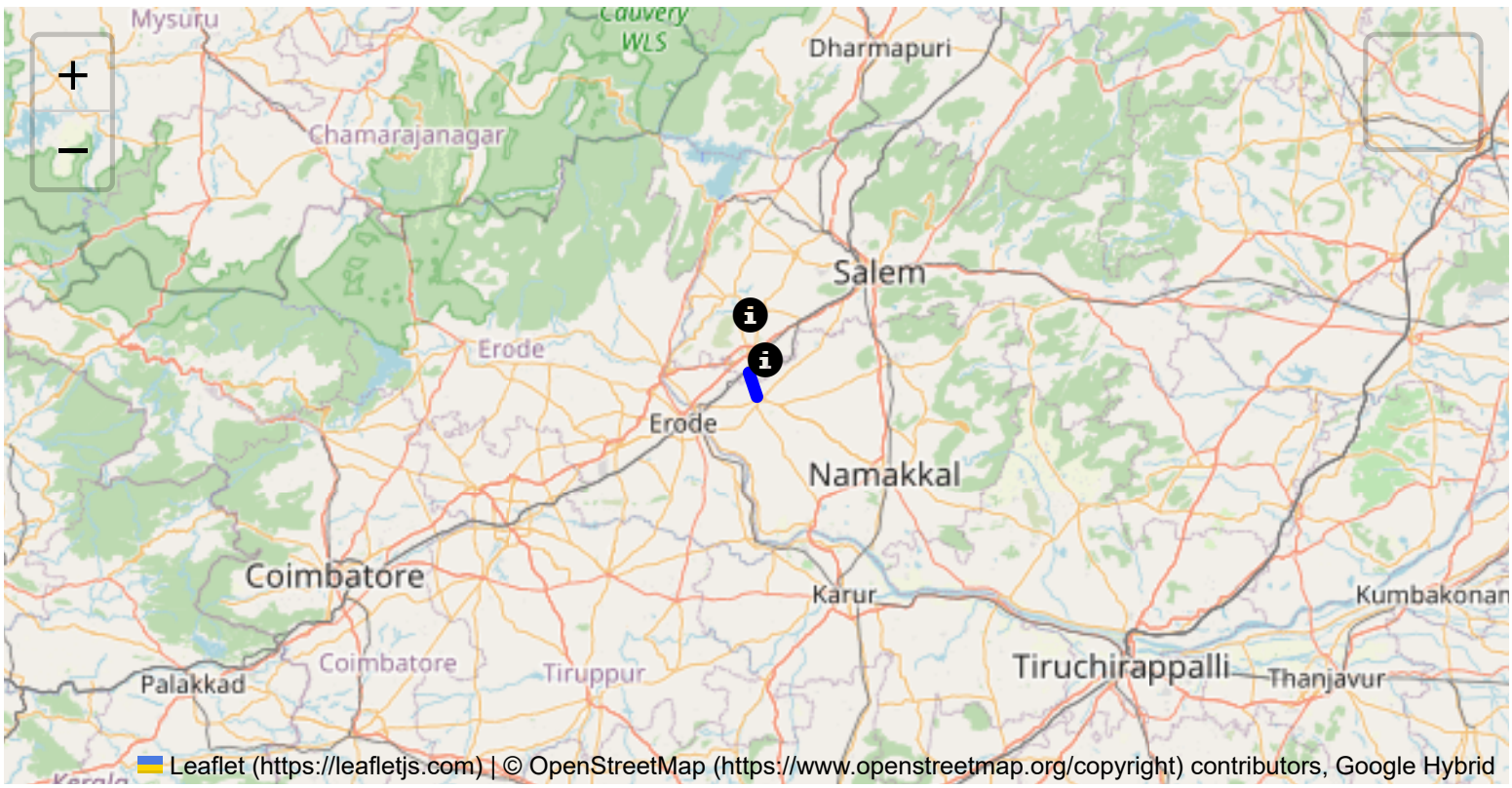
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

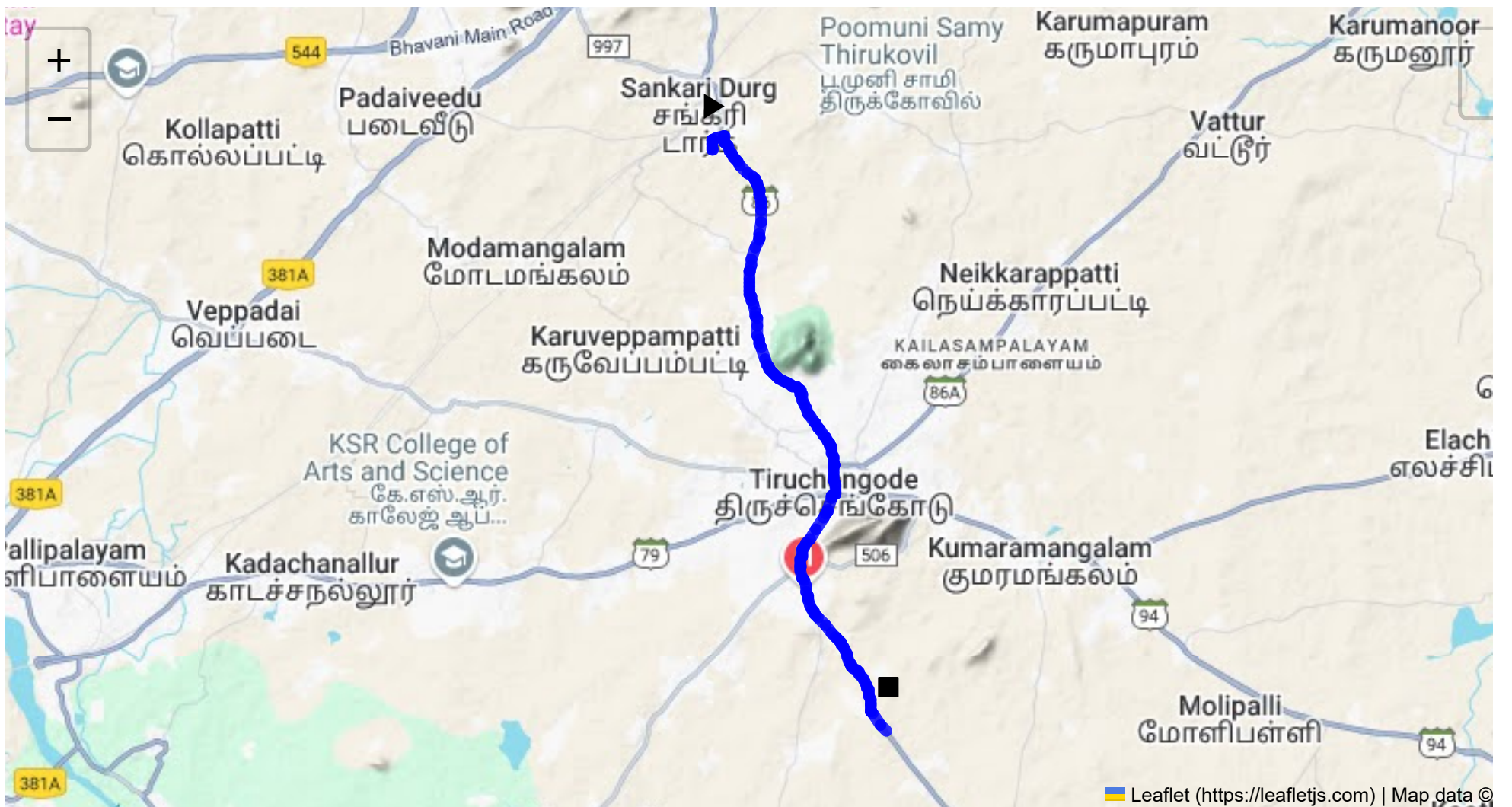
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 13.31 km
Estimated Duration: 0.4 hours
Adjusted Duration (Heavy Vehicle): 0.5 hours
Start: (11.4381, 77.8734)
End: (11.337913, 77.904083)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route from Sangagiri to Puduppuliyampatti, Tamil Nadu, spans approximately 13.31 kilometers. The path typically traverses local roads through a mix of rural and semi-urban environments. The detailed

route typically focuses on direct local roads to ensure efficiency while maintaining the security of transport.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region generally experiences a tropical climate with the following major seasons:

- **Summer (March to June):** Hot and humid, with temperatures often exceeding 35°C.
- **Monsoon (June to September):** Heavy rainfall typically leads to flooding and waterlogging on roads.
- **Winter (December to February):** Mild weather with less risk of weather-related disruptions.

Potential weather-related hazards include reduced visibility during heavy rains and possible flooding in low-lying areas during monsoons.

3. Analysis of Traffic Patterns

Traffic patterns along this route are generally less congested compared to urban centers. However, peak hours in the morning (7:30 - 9:30 AM) and evening (5:30 - 7:30 PM) may cause delays due to school and work commute traffic, particularly near market areas and school zones.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route. Key observations include:

- Paved roads in major segments, though some sections may have potholes or uneven surfaces.
- Limited signage and narrow lanes in rural parts require careful driving.
- Occasional roadwork can lead to temporary detours.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, alternative routes may involve taking the nearest state highways for faster access to major centers with better infrastructure. Local maps and navigation tools should be checked real-time to find the shortest alternative, considering traffic and road conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has specific regulations on the transport of hazardous materials. These include:

- Mandatory permits for transport.
- Restriction from residential areas and proximity to schools or hospitals.
- Designated routes for hazardous material where available.

7. Overview of Historical Incidents

There have been sporadic reports of road accidents involving heavy vehicles predominantly due to monsoon-related factors such as slippery roads. No significant past incidents involving hazardous materials have been recorded near this route.

8. Environmental Considerations and Sensitive Areas

The area is dotted with agricultural fields and small villages. Transporters must be careful to avoid spills or leaks to prevent environmental damage, particularly near water bodies or farmlands.

9. Analysis of Communication Coverage

The route generally has good mobile network coverage, but potential dead zones may occur in remote rural sections. Ensuring communication devices are fully charged and alternative contacts are available is recommended.

10. Estimated Emergency Response Times

Emergency response times might be delayed in rural areas with estimates of 30-45 minutes due to the distance from major towns. Urban areas would typically have faster response times of 15-30 minutes.

11. Overall Summary of Risk Assessment

- Low to Moderate Risk:** The route is generally safe with precautions regarding weather and road conditions.
- Critical Concerns:** Monsoons can severely affect driving conditions due to road flooding and low visibility.
- Safety Measures:** Ensuring vehicle preparedness, having an emergency plan, and close monitoring of weather forecasts are crucial.

The route from Sangagiri to Puduppuliyampatti is operationally viable with planned contingencies, adherence to regulations, and effective risk management strategies.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
6	Turn	High	11.37850, 77.89453	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453

